AI for SARS

Shared Understanding and Painpoints

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DAY I

Content

What is AI, What is Not.

Content

Content

Content

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Introduction — Artificial Intelligence

- “*Machines with Minds*”

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- **Goals:** NLP, Robotics, Planning, Machine Learning, Knowledge Reasoning and/or Thinking, Computer Vision, Problem Solving, Game Playing, etc.
- **Applications:** Education, Business, Economics, Society, Healthcare, Government, Warfare, etc.

Techniques (Statistical vs Rule-based)

- **Statistical**
  - ML Models (MLP/ANN, CNN, etc.) built on Good Data
  - Existing Models / APIs with New Applications

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- **Symbolic / Rule-based**
  - Formal Techniques
  - Deterministic vs Probabilistic Rules
  - Context-free / context-sensitive formal languages

Spectrum of Possibilities

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Level 1 — Simple Technologies

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- Cloud and its Technologies : Online, you can draw, translate, transcribe, check sentiments of social media texts, etc.

Level 2 — Non-technical Topics and Issues around 4IR

- Ethics of 4IR Techs, Dangers of 4IR
- Social Impact of AI
- AI and Politics, Cyberwarfare
- AI and Governance
- Adoption, Conspiracies and Social Media
- Access to 4IR
- Data Bias
- Language bias / Under-resourced languages
- Terrorism and 4IR
- New Frameworks for Regulation

Level 3 — Research Methods, Software Tools, and New Data Sources

- New Data Sources (e.g. Google Datasets)
- Tools (e.g. Matlab, SPSS, RStudio, etc.)
- Research methods
  - Prediction Models : Logistics Regression, etc.
  - Data Mining and Visualisation
  - New Algorithms or Models
  - Digital Marketing, Social Media Analytics
  - Serious Games / Educational Games
- Big Data Sources
  - Social Media
  - Machines / Hardware

Level 4 — Off-the-shelves, APIs and Applications

- Chatbots (e.g. Chat-GPT)
- VR/AR (Dreams replaying, therapy VR, VR edu, VR entertainment, etc.)
- 3D Printing, Gesture-base Computing
- Blockchain and Cryptocurrency
- Apps for Deepfakes
- IoT
- Off-the-shelve Robots and Drones (e.g. Pepper, Photo-drones, etc.)
- Drones in Agriculture
- Voice AI Systems : Siri / Alexa
- AI Apps across Domains (Finance, Accounting, etc)

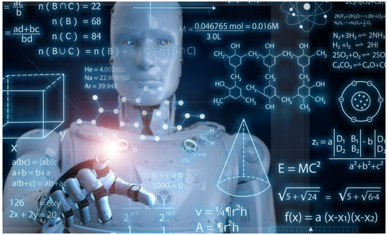
Level 5 — Data Science and Applications

- Big Data and Data Science
- Supervised Learning
- Nanotechnology
- Bio-Statistics
- Prediction : Earthquake, wealther, and other applications

SARS

Level 6 — Machine [Deep] Learning, New Algorithms and Tools

- Unsupervised and Deep Learning
- Quantum Computing
- Deep-fake (Detection)
- Surveillance : Analysis of Real-time Videos

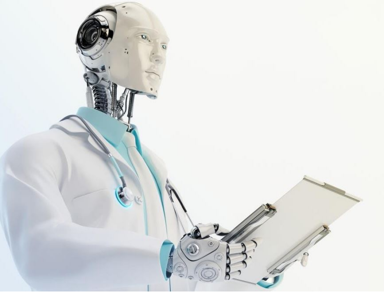


Level 7 — Expert Tools

- Game Playing AIs (e.g. Deep Blue)
- Robotic Surgery

Level 8 — Multidisciplinary Computation and Special Applications

- Cure/Treatment for Cancer
- Global Warming (Weather Control), etc



Level 9 — Advanced Drones and Machine

- War machines, military drones,
- Autonomous vehicles (DARPA, space navigation)
- Flying man
- Hyper-fast trains
- Space travel and exploration

Level 10 — The Dark Hole : Gene Editing and Super-soldiers

- Bio-weapons
- Genetic Modification of Babies
- Gene Editing, Edited/Super Soldiers
- Bio-Printing

Game — Speaking AI!

Demonstrations — The Showroom

Build Session

Build Session (Closed)

AI for SARS: Perception and Painpoints

SARS Technology Landscape (1/4) — Core Tax & Digital Systems

System	Function	Year	Built / Bought
eFiling	Online tax submission platform	2001	Built (Hybrid)
SARS MobiApp	Mobile tax services	2014	Built
Revenue Accounting System	Core tax revenue accounting	Early 2000s	Bought
Integrated Tax System	Legacy tax admin platform	1990s	Bought

SARS Technology Landscape (2/4) — Customs & Workflow Systems

System	Function	Year	Built / Bought
Modernised Customs System	Trade declarations processing	2013–2018	Hybrid
Customs Declaration Processing	Import/export clearance	Pre-2010	Bought
Case Management System	Audit	enforcement tracking	2010
Hybrid			
Debt Management System	Tax debt recovery	2012	Hybrid

SARS Technology Landscape (3/4) — Data & Decision Intelligence

System	Function	Year	Built / Bought
Enterprise Data Warehouse	Central analytics platform	2004	Bought
Risk Engine	Compliance risk scoring	2006–2010	Built
Automatic Assessment Engine	AI-based return assessment	2020	Built
Third-Party Data Platform	Bank & employer data ingestion	2008+	Built

SARS Technology Landscape (4/4) — Enterprise & Infrastructure

System	Function	Year	Built / Bought
SAP ERP	Finance & HR systems	Mid-2000s	Bought
Enterprise Content Management	Digital records	2015	Bought
API Integration Layer	System interoperability	2018–2022	Built
Cybersecurity Stack Bought	Threat detection	SOC tools	Ongoing

Recent AI Initiatives at SARS (1/2)

- **Automatic Assessments (2020–)**
  AI-based return pre-population and low-risk auto-finalisation.

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  Predictive audit targeting and anomaly detection.

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- **Advanced Risk & Compliance Analytics**
  Predictive audit targeting and anomaly detection.

- **Illicit Economy Detection**
  Network analytics for syndicates and trade fraud.

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- **Digital Behavioural Analytics**
  Login anomaly detection and fraud pattern monitoring.

Major SARS Pain Points (1/3)

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  Continuous exposure to attacks, identity theft, refund fraud.

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- **Changing Tax Legislation**
  Frequent legal updates increase system and training burden.

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  User errors increase support demand.
- **High Query Volumes**
  Backlogs in disputes, audits, and verifications.

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  Scarcity of advanced data expertise.

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- **Public Trust & Reputation**
  Service friction affects institutional confidence.

Discussions & Reflections on Day I

Describe Day I — a word or phrase (Closed)

- Use one word, a phrase, or a short sentence to describe Day I.

Thank You — Please Let's Connect

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